Not up to its critical perspective on digitalization: A Descriptive Analysis of How Sustainability is Approached in the ICT4S Conference

Clément Morand[1,*,†], Aina Rasoldier[2,†] and Paul Gay[3,†]

[1]Université Paris-Saclay, CNRS, LISN, Orsay, France
[2]Univ. Grenoble Alpes, CNRS, Grenoble INP, G2Elab, Grenoble, France
[3]Université CY, laboratoire ETIS, Pau, France

Abstract

The Information and Communication Technologies for Sustainability (ICT4S) conference gathers a diverse community of players. Among those different players, different conceptions of sustainability, its links with Information and Communication Technology (ICT), and actions toward sustainability exist. Controversies may arise around certain aspects of the presented research. This study aims to better understand what the ICT4S community values by examining publications at the conference. We explore whether different visions of sustainability and links between ICT and sustainability are presented at the conference. We perform a thematic analysis of the proceedings of the 2022 and 2023 editions of the conference. We find that the contributions to ICT4S are mostly focused on greenhouse gas emissions, with research primarily targeted at other researchers and digital systems designers, with a strong separation between work on sustainability by ICT and the sustainability of ICT. The ICT4S community seems to overall still comply with the initial vision of ICT4S as expressed in its 2013 manifesto. This analysis aims to give members of the ICT4S community a better understanding of the community and to foster dialogue on what collective sustainability vision the conference could support. To achieve its critical perspective on digitalisation promises, we think the ICT4S conference should value more works aiming at transformative changes with a more holistic vision of sustainability, and better integration of ICT sustainability in works leveraging ICT toward sustainability.

Keywords

ICT4S community, thematic analysis, sustainability

1. Context and motivation

The *ICT for Sustainability* (ICT4S) conference emerged in 2013 with a unique vision, described as a critical perspective on *Information and Communication Technologies* (ICT). This perspective, and the subsequent research community, was summarized in a document that can be interpreted as a manifesto for the ICT4S community. It states, "only with targeted efforts on the part of politics, industry, and consumers will it be possible to unleash the true potential of ICT to create a more sustainable society." [1, p. 284]. Such efforts include sustainability *in ICT*, *by ICT*, ICT to create incentives for sustainable lifestyles, and research and education on ICT4S. Even though a theoretical consensus exists on the vision of the ICT4S community controversies still plague the conference.

Among the controversies on the role and place of ICT in sustainability, its carbon footprint and the speed of its growth are debated [2, 3]. This was illustrated during the 2023 edition of the conference in Rennes, where a vivid debate on the exactitude of figures on the environmental impact of ICT and its evolution followed one of the keynote speeches.

How can the divergences that traverse the community be explained when a shared vision should exist? According to Rugeviciute et al. [4], different practices, visions of sustainability and controversies

ICT4S 2026

*Corresponding author.
†These authors contributed equally.
clement.morand@lisn.upsaclay.fr (C. Morand); aina.rasoldier@univ-grenoble-alpes.fr (A. Rasoldier); paul.gay@univ-pau.fr (P. Gay)
0009-0007-8575-9874 (C. Morand)

© 2026 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

can arise because people consider different time horizons, ethical values, and potentials of technology. de Goër and Quinton [5] showed that views on sustainability are rarely justified within the ICT4S publications. We thus need to better understand what is valued regarding sustainability in the ICT4S community. This could help to point out to tensions that could spark controversies.

To that end, we perform a qualitative analysis on a corpus composed of the manuscripts published on the proceedings of the 2022 and 2023 editions of the conference. As Birhane et al. [6] observe, research papers provide valuable material for understanding what a community values most. Indeed, manuscripts are partly written to be approved by reviewers and to obtain approval from other researchers through citations.

Our contributions include a fine-grained description of the papers presented at the ICT4S conference at the 2022 and 2023 editions, insights on the different visions of sustainability described, and how they correlate with scientific context and methodology.

2. Methodology

This research project was conducted by three computer scientists who were working on the environmental impacts of ICT. The 39 articles published during the 2022 and 2023 editions of ICT4S were *thematically analysed* [7], as these were the two most recent years for which proceedings were available when carrying out the analysis. The analysis was conducted based on an *analytical grid* derived from our research questions and refined after a first reading of three articles from the 2022 proceedings. The grid is structured around two topics: (1) the scientific positioning of the paper and (2) the vision of sustainability presented in the manuscript.

Annotation was performed in three phases. First, each author annotated an equal share of the manuscripts from 2022. To ensure consistent annotation practices each pair of authors annotated a paper in common and regular meetings were held throughout the process. Each author structured their analysis as a *thematic tree*, (i.e., a hierarchical structure that organizes the annotated themes in (possibly nested) categories), which were then merged to created a unified analysis of the 2022 edition, triangulating the analysis with each individual sensitivity. Themes were mostly added inductively, but also sometimes speculatively and then confirmed or removed based on if they turned out to appear or not in the rest of the corpus. This process was repeated on the proceedings of the 2023 edition, using mainly themes derived at the previous step to consolidate the analysis and incorporate emerging aspects.

Third, the reported list of themes for each manuscript was normalized according to the tree. The transcripts of themes for each of the 6 manuscripts annotated by two authors were then used to compute inter-annotator agreement, using a variant of Fleiss' κ [8], considering the presence or absence of each possible theme in each transcript. We obtain an agreement per theme, which we average to obtain a global score of $\simeq 0.56$, which can be interpreted as a *moderate agreement* [9, §1.5.2, p. 41–43]. After discussing the inter-annotator agreement and creating a consensus on the annotation for each of the six annotated manuscripts, we revised all annotations to ensure that they aligned with the final consensus.

Finally, we also produced descriptive statistics of the corpus, based on the revised annotations. These were notably used to assess possible correlations between the different pairs of themes.

Second, we discuss threats to validity coming from the quality of annotations. We got a moderate agreement, according to the inter-annotator score, which implies that the correlation coefficient we compute might differ slightly if we were to re-annotate our corpus. To mitigate the impact of our relatively low agreement score on the pertinence of derived correlations, we reviewed our annotations in light of the consensus we found on the inter-annotator agreement corpus. The main correlations we discussed are still relevant as we obtain those on themes that have a relatively good agreement, namely when we chose that a paper is addressing sustainability by ICT or sustainability in ICT.

3. Results

Climate change is considered as the most important sustainability issue in a sustainable development framework. More than half of the manuscripts, focus on indicators related to climate change (i.e., greenhouse gas (GHG) emissions, energy usage, or carbon footprint of products).

Alongside this focus on climate change as a primary sustainability issue, sustainability is most often approached under the framing of sustainable development, with its three pillars (environmental, social, and economic) sometimes framed with the United Nations' Sustainable Development Goals.

Still, The ICT4S community presents an interesting variety of visions on sustainability with some works aligned with the concept of degrowth, which is incompatible with visions aligned with the concept of decoupling. As mentioned in the introduction, visions of sustainability can diverge because of different assumptions about relevant time horizons, and ICT potential. We found that papers advocating systemic changes and ICT degrowth highlight the importance of discussing assumptions on the pathways toward sustainability and the role ICT plays in desirable futures. On the other hand, they are rarely discussed in papers which use the main framing of sustainable development.

Research mostly targeted at researchers and designers and working on individual scale actions. A structuring choice to achieve sustainability is to decide who is the target audience of the paper and at which scale and to which actor the research proposal applies. We found that the recommendation and tools described in the paper are produced for researchers, companies, policy makers and citizens. We also characterised the different scales at which the sustainability vision is applied : individual, community, economic domain, professional activity or world wide scales.We found that the most frequent target is researchers or designers, and scientists and designers are actors of sustainability in around half of the papers. We categorize around half of the manuscripts as working on individual-scale action (behavior of consumers or ICT users). Apart from a few exceptions, we did not find research co-design. A second finding is that the scale is correlated with research practices. Around two-thirds of the environmental assessments are performed at a micro-economic scale or only assess one system [e.g., 10, 11]. Doing an environmental assessment at a micro-scale often implies the consideration of levers at individual scale (we find a positive correlation of 0.42). That could be explained by the fact that it is complex to get measurement data on large systems.

Works on sustainability by ICT often do not consider sustainability in ICT. Finally, we denote the links between ICT and sustainability the publications. We observe that, for almost the entire corpus, links between ICT and sustainability either fall within the category of sustainability in ICT (i.e., the environmental impact of digital infrastructure) or of sustainability by ICT (i.e., mobilizing ICT for impact reduction). Those two categories represent approximately the same volume of articles. The separation of these two groups is characterized by a negative correlation coefficient (-0.46). In other words, our analysis indicates that papers on sustainability by ICT often do not consider sustainability within ICT.

4. Conclusion

As manuscripts do not necessarily reflect the values of their authors, our study should not be interpreted as an analysis of the values of researchers of the ICT4S community It however gives valuable insight into what is valued by the community as a whole. We thrived to describe differences in scientific positioning or vision of a sustainability and the role and place ICT plays in sustainability, based on an analysis of manuscripts presented at the ICT4S conference.

Our analysis shows that the community's vision still overall complies with the manifesto written after the first edition of the conference. The most frequent positioning in papers focuses on assessing GHG emissions and aims at technological levers and modifying practices through eco-design. The ICT4S community is also interdisciplinary, welcoming numerous methodologies coming from the humanities.

Two distinct groups seem to exist within the community, focusing either on sustainability *in* ICT or on sustainability *by* ICT, with few projects linking the two aspects. Our analysis might be valuable for newcomers, enabling them to better understand the state and orientations of the ICT4S community.

We agree with Mann et al. [12] that most of the research presented at the conference focuses on small and incremental changes, with only a small proportion of articles trying to elicit more transformational change. The central vision of sustainability presented at the conference focuses solely on environmental sustainability, especially GHG emissions. We would welcome more studies incorporating more social components of sustainability, especially the relations between social and environmental aspects.

Our analysis indicates that papers working on sustainability *by* ICT often fail to consider sustainability *in* ICT. We think these two aspects would benefit from being studied in coordination — both to ensure that proposals for sustainability *by* ICT are not voided by possible un-sustainability *in* ICT and that they fit within a desirable future.

References

[1] L. M. Hilty, CONFERENCE RECOMMENDATIONS: HOW to IMPROVE the CONTRIBUTION of ICT to SUSTAINABILITY, in: L. M. Hilty, B. Aebischer, G. Andersson, W. Lohmann (Eds.), Proceedings of the first international conference on Information and Communication Technologies for Sustainability, ETH Zurich, Switzerland, 2013, pp. 283–287. doi:10.3929/ethz-a-007337628.

[2] C. Freitag, M. Berners-Lee, K. Widdicks, B. Knowles, G. S. Blair, A. Friday, The real climate and transformative impact of ICT: A critique of estimates, trends, and regulations, Patterns 2 (2021) 100340. doi:10.1016/j.patter.2021.100340.

[3] A. Pasek, H. Vaughan, N. Starosielski, The world wide web of carbon: Toward a relational footprinting of information and communications technology's climate impacts, Big Data & Society 10 (2023) 20539517231158994. doi:10.1177/20539517231158994.

[4] A. Rugeviciute, V. Courboulay, L. M. Hilty, The Research Landscape of ICT for Sustainability: Harnessing Digital Technology for Sustainable Development, in: 2023 International Conference on ICT for Sustainability (ICT4S), 2023, pp. 97–107. doi:10.1109/ICT4S58814.2023.00019.

[5] B. de Goër, S. Quinton, The Political Dimension of Problematizing ICT-related Sustainability Issues, in: ICT4S 2025 - International Conference on Information and Communications Technology for Sustainability, Dublin, Ireland, 2025, pp. 1–12. URL: https://inria.hal.science/hal-05064545.

[6] A. Birhane, P. Kalluri, D. Card, W. Agnew, R. Dotan, M. Bao, The values encoded in Machine Learning research, in: Proceedings of the 2022 ACM conference on fairness, accountability, and transparency, FAccT '22, Association for Computing Machinery, New York, NY, USA, 2022, pp. 173–184. doi:10.1145/3531146.3533083.

[7] P. Paillé, A. Mucchielli, L'analyse qualitative en sciences humaines et sociales - 5e éd., Armand Colin, Paris, 2021.

[8] J. L. Fleiss, Measuring nominal scale agreement among many raters, Psychological Bulletin 76 (1971) 378–382. doi:10.1037/h0031619.

[9] A. Baledent, De la complexité de l'annotation manuelle : méthodologie, biais et recommandations, Ph.D. thesis, Normandie Université, 2022.

[10] J. Malmodin, Just measure IT! - Electricity consumption measurements of electronic devices and estimates of datacenter and network services for one household, in: 2023 International Conference on ICT for Sustainability (ICT4S), 2023, pp. 35–45. doi:10.1109/ICT4S58814.2023.00013.

[11] D. Carastan-Santos, K. Rzadca, L. Sousa, D. Trystram, Social and Environmental Effects of Post-COVID-19 Computer Science Virtual Conferencing: The Euro-Par Case, in: 2023 International Conference on ICT for Sustainability (ICT4S), 2023, pp. 132–141. doi:10.1109/ICT4S58814.2023.00022.

[12] S. Mann, O. Bates, R. Maher, Shifting the maturity needle of ICT for Sustainability., in: ICT4S, 2018, pp. 209–226.